Programming In C
Final Project

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Project Overview

This project is based on a mini project where I used an existing C program that simulates a Rock, Paper, Scissors game. The purpose of this final project was to improve that program by adding new features, fixing bugs, and making the code easier to read and maintain. The focus was on thinking like a developer by refining and improving the existing piece of software, rather than creating something new.

Project overview

Original Program Issues

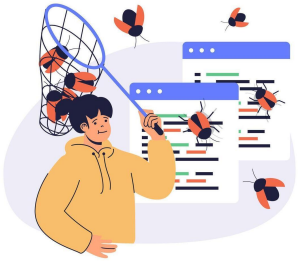
The original program was very basic and had several limitations. It only allowed the user to play one round, and there was no option to replay the game. It also did not validate user input, meaning incorrect inputs could cause unexpected behavior. Additionally, the code structure was not well organized, with most logic written in one place, making it harder to read and maintain. There were also small bugs, such as missing return values and issues with random number generation.

Improvements Made

Several improvements were made to enhance both the functionality and quality of the program. Input validation was added to ensure that only valid choices are accepted, improving reliability. A game loop was also implemented to allow multiple rounds of play, making the program more interactive. The code was also restructured into multiple functions, improving readability and organization. In addition, a score tracking system was added to keep track of wins, making the game feel more complete.

Debugging Examples

During the improvement process, several issues were identified and fixed. One major issue was a missing return statement in the game logic, which could cause unpredictable behavior. Another issue involved incorrect random number conditions that did not cover all possible values. There was also a problem with user input handling due to how ‘scanf’ reads characters, which was fixed by adjusting the input format. Finally, the lack of input validation was addressed by adding a loop that ensures correct input before continuing.



Program Demo Screenshots

The improved program was tested using a variety of inputs to ensure stability and correctness. Normal inputs were used to verify that the game logic works as expected, while invalid inputs were tested to confirm that the validation system responds properly. Multiple rounds were also tested to ensure the loop and the score tracking function correctly. Screenshots included in this presentation demonstrate these test cases and show the program running successfully.

Game Start

```
PS C:\Users\Owner\Desktop\School Stuff\Year 1 Semester 2\Programming In C\FINAL PROJECT> .\game
=== Stone Paper Scissors Game ===

Enter your choice:
s = Stone
p = Paper
z = Scissors
Your choice: s

You chose: s | Computer chose: z
You Win!

Score -> You: 1 | Computer: 0

Do you want to play again? (y/n):
```

Lose

```
PROBLEMS   OUTPUT   DEBUG CONSOLE   TERM

Score -> You: 1 | Computer: 0

Do you want to play again? (y/n): y

Enter your choice:
s = Stone
p = Paper
z = Scissors
Your choice: s

You chose: s | Computer chose: p
You Lose!

Score -> You: 1 | Computer: 1

Do you want to play again? (y/n):
```

Draw

```
PROBLEMS   OUTPUT   DEBUG CONSOLE   TE

z = Scissors
Your choice: t
Invalid input! Please try again.

Enter your choice:
s = Stone
p = Paper
z = Scissors
Your choice: s

You chose: s | Computer chose: s
Game Draw!

Score -> You: 1 | Computer: 1

Do you want to play again? (y/n):
```

Invalid Input

```
PROBLEMS   OUTPUT   DEBUG CONSOLE   TE

Score -> You: 1 | Computer: 1

Do you want to play again? (y/n): y

Enter your choice:
s = Stone
p = Paper
z = Scissors
Your choice: t
Invalid input! Please try again.

Enter your choice:
s = Stone
p = Paper
z = Scissors
Your choice:
```

Win

```
Enter your choice:
s = Stone
p = Paper
z = Scissors
Your choice: s

You chose: s | Computer chose: s
Game Draw!

Score -> You: 1 | Computer: 1

Do you want to play again? (y/n): n

Final Score -> You: 1 | Computer: 1
Thanks for playing!
PS C:\Users\Owner\Desktop\School Stuff\Y
```

What I Learned

This project helped me understand the importance of improving existing code rather than just writing new programs. I learned how to debug issues, handle user input more effectively, and structure code using functions to make it more readable and maintainable. It also showed me how small improvements, such as validation and better organization, can significantly improve a program’s quality and user experience. Overall, this project helped me develop a more professional approach to programming.